

Gabriel de Aragão Aleksandravicius

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Technologies

Languages: JavaScript, TypeScript, C/C++, PHP, Go, SQL, Python, HTML5/CSS3

Technologies: Node.js, ReactJS, VueJS, Next.js, Tailwind CSS, PostgreSQL, MongoDB, LangChain, Redis, WebSockets, MQTT, Express.js

Tools: Linux, Docker, Git, Keycloak, Gitlab CI, GitHub Actions

Practices: Domain-Driven Design (DDD), Test-Driven Development, CI/CD, Agile, REST API Design, Microservices, Event-Driven Architecture

Experience

Sentosa Technology Consultants

Full Stack Software Engineer / Embedded Software Engineer

April 2024 – Present

Remote (Bern, CH)

- **Quality By Design / Network Side Probing**

- Embedded software engineer working on a research project that aims to guarantee network quality for end users by detecting degradation and automatically triggering network-level interventions.
- Developing the measurement layer directly on the home router: detecting which application servers a user is communicating with and passively measuring connection quality.

- **CMMC Compliance Automation Platform (AI-Powered)**

- Backend developer responsible for architectural design and system integration; built a RESTful API in Node.js and TypeScript, designed a normalized PostgreSQL schema, and integrated Keycloak and CASL for authentication and role-based access control.
- Developed inter-service communication with the AI engine via secure webhooks and external API triggers.

- **STAMP Protocol Latency Testing Platform:** Designed a web app and backend services using socket programming for STAMP packet generation, transmission, and precise timing.

CERN

Full Stack Software Developer

Jan 2021 – Mar 2024

Geneva, CH

- Developed and maintained Glance, the ATLAS collaboration's management system for personnel, speaker selection, and publication workflows.
- Built REST APIs in PHP with Domain-Driven Design to support VueJS frontends; modernized legacy code through incremental refactoring.
- Led integration of ATLAS member registration with CERN HR API, automating workflows and saving up to 1 hour daily for the Secretariat.
- Automated dynamic synchronization between Glance and CERN HR databases, reducing manual work by 30 minutes daily.
- Dockerized Glance database for consistent local development; introduced ephemeral databases and Liquibase migrations in GitLab CI/CD to enhance testing and deployment.

UFRJ Aerospace Research Group

Avionics Engineer / Manager

Sep 2019 – Dec 2021

Rio de Janeiro, Brazil

- Led a team of 4 in developing a PC/104 flight computer for data acquisition, telemetry, and parachute deployment, using Altium Designer for hardware and embedded C/C++ for software.
- Avionics system deployed in the 2023 Latin American Space Challenge: 1st place in the 3 km Solid Motors category; 2nd place in 2020 in both Solid Motors and Hybrid/Liquid categories.

Education

Haute École Arc (HE-Arc) - Neuchâtel

- Bachelor of Science in Computer Science, Major in Embedded Software

Sep 2025 (ongoing)

University of Rio de Janeiro

- Bachelor: Bionanotechnology

Mar 2018 – Dec 2021 (discontinued)

Languages

English/Portuguese (C2), French (C1), German (A1)

Publications

Enhancing data consistency in ATLAS and CERN HR databases through automated synchronization

May 2023

Gabriel de Aragão Aleksandravicius et al.

<https://doi.org/10.1051/epjconf/202429505004>

Additional Information

- **CHEP 2023** (Norfolk, USA): Talk at the 26th International Conference on Computing in High Energy Nuclear Physics on automated data synchronization between ATLAS Glance and CERN HR databases.
- **ATLAS Speakers Committee Workshop** (CERN, 2022): Presented ongoing and planned developments in ATLAS Glance tooling.